PES UNIVERSITY • DEPT. OF COMPUTER SCIENCE AND ENGINEERING

Lab 2: Epics & User Stories Specification

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Problem Statement: #43 – Feature Flag & Dynamic Config Manager

Course: Software Engineering Laboratory (Lab 2)

1. Epics Architecture

Epic ID	Summary	Scope	Traceability
FFM-EPIC-1	Flag Lifecycle & Metadata	Flag CRUD, metadata, multi-env overrides	FR-002, FR-004, UC-01, UC-04
FFM-EPIC-2	Targeting & Gradual Rollouts	Deterministic hashing, attribute targeting	FR-001, FR-003, UC-02, UC-03
FFM-EPIC-3	Governance & Audit Logging	Approval gate, visual diff, audit logging	UC-06, UC-07, NFR-002
FFM-EPIC-4	Evaluation Engine & SDK Sync	Sub-15ms p95, real-time WebSocket sync	FR-005, NFR-001, UC-05, UC-03

2. Product Backlog: User Stories Catalog

ID	Epic	User Story Statement	Priority	Pts	Sprint	Acceptance Criteria
US-01	FFM-EPIC-1	As a Software Engineer, I want to create a feature flag with a unique key, description, and default state, so that client apps can reference the flag unambiguously.	HIGH	3	Sprint 1	Pass: Duplicate key triggers 409 conflict; unique key registers < 2s. Fail: Duplicate keys accepted.
US-02	FFM-EPIC-2	As a Software Engineer, I want to configure a percentage rollout (0-100%) based on a deterministic hash of user IDs, so that I can gradually roll out features and limit blast radius.	HIGH	5	Sprint 1	Pass: 25% rollout allocates 25% (±2%) of 10,000 simulated IDs; hash mapping is strictly deterministic. Fail: User changes cohort across evaluations.
US-03	FFM-EPIC-2	As a Software Engineer, I want to evaluate user targeting rules for specific cohorts, so that targeted users receive early feature access.	HIGH	5	Sprint 1	Pass: User matching targeted attribute receives true; non-targeted user receives fallback. Fail: Non-targeted user receives feature.
US-04	FFM-EPIC-4	As a Software Engineer, I want the evaluation API to execute in under 15ms, so that embedding flag checks into critical paths adds negligible latency.	HIGH	8	Sprint 1	Pass: p95 latency under simulated peak load is < 15ms; in-memory cache lookup avoids disk I/O. Fail: Evaluation p95 exceeds 15ms.
US-05	FFM-EPIC-4	As a Client SDK Developer, I want the SDK to maintain a local in-memory cache with fallback defaults, so that network partitions do not break host application features.	MEDIUM	3	Sprint 1	Pass: In-memory evaluations return in < 1ms during offline conditions; graceful fallback. Fail: Application blocks on network drop.

ID	Epic	User Story Statement	Priority	Pts	Sprint	Acceptance Criteria
US-06	FFM-EPIC-1	As a Software Engineer, I want to define environment-specific overrides (Dev, Staging, Prod), so that I can test flags in lower environments without exposing code in Prod.	MEDIUM	5	Sprint 2	Pass: Prod override set to OFF while Staging is ON returns false for Prod and true for Staging. Fail: Overrides leak between environments.
US-07	FFM-EPIC-3	As a Software Engineer, I want to save changes as pending drafts and preview a visual diff, so that I can verify proposed modifications before requesting approval.	MEDIUM	3	Sprint 2	Pass: Draft changes do not alter runtime configurations; diff clearly highlights altered rules. Fail: Draft changes alter live evaluations.
US-08	FFM-EPIC-3	As a Release Manager, I want to review, approve, or reject pending flag changes with validation checks, so that invalid configs never reach production.	HIGH	5	Sprint 2	Pass: Only authenticated Release Managers can publish; invalid rules (rollout >100%) rejected. Fail: Unauthorized user publishes changes.
US-09	FFM-EPIC-3	As a Release Manager, I want all configuration changes logged in an immutable audit trail with one-click rollback, so that actions are auditable and regressions revert instantly.	HIGH	5	Sprint 2	Pass: Tamper-proof audit record created for every publish; one-click rollback restores prior state < 3s. Fail: Change occurs without audit entry.
US-10	FFM-EPIC-4	As a Client SDK Developer, I want updated flag states streamed via WebSockets in real time, so that clients update within 5 seconds without redeployment.	HIGH	5	Sprint 2	Pass: 95% of active SDK connections receive updates within 5s; automatic reconnect upon drop. Fail: Clients take > 30s to synchronize.

3. Planning Poker Estimation Rationale

Story points follow the Fibonacci sequence (`1, 2, 3, 5, 8, 13`), calibrating effort and risk:

- **3 Points:** Standard CRUD models and UI rendering (`US-01`, `US-05`, `US-07`).
- **5 Points:** Algorithmic hashing, multi-tenant state isolation, WebSocket channel management, and audit records (`US-02`, `US-03`, `US-06`, `US-08`, `US-09`, `US-10`).
- **8 Points:** Critical-path optimizations with high uncertainty (`US-04`: Sub-15ms p95 latency benchmark).